

Experiment No.: 6

- 1. Title:** Use string functions and regex for email validation, data extraction, and text analysis.
- 2. Software/Tools Required:** Visual Studio Code (VS Code), web browser, HTML, JavaScript.

3. Theory: String functions and Regular Expressions (Regex) are used in JavaScript to process and analyze text. String functions such as `toLowerCase()`, `toUpperCase()`, `includes()`, `split()`, `trim()`, and `replace()` help manipulate and search text. Regular Expressions are patterns used to find specific text formats, such as validating an email address, extracting numbers or words, and identifying patterns in a sentence. In a basic HTML, CSS, and JavaScript application, HTML provides the input fields and structure, CSS provides styling, and JavaScript uses string functions and Regex to validate emails, extract required data, and perform simple text analysis.

4. Output :

127.0.0.1:5500/pract06.html1189Rutva

Email Validation, Data Extraction & Text Analysis

Enter your text:

```
Hello, my name is Rutva Sawarkar.  
My email addresses are rutva@gmail.com and rutva123@yahoo.com.  
You can contact me at 9876543210.  
I am a B.Tech CSE student learning Python, Data Analysis and Machine Learning.  
Python is easy to learn and Python is very useful for data analysis.  
Regular expressions are useful for email validation and text processing.
```

Analyze Text

1. Email Validation

rutva@gmail.com → **Valid**
rutva123@yahoo.com → **Valid**

2. Phone Number Extraction

9876543210

3. Text Analysis

Total Characters: **351**
Total Words: **55**
Unique Words: **40**

4. Word Frequency

Frequency of "python": 3

5. Words Starting With 'P'

python python python processing

6. Words Ending With 'ING'

learning learning processing

7. Words Longer Than 6 Characterssawarkar addresses rutvagmailcom rutva123yahoom contact 9876543210 student learning
analysis machine learning analysis regular expressions validation processing**8. String Functions****Uppercase Text:**

HELLO, MY NAME IS RUTVA SAWARKAR. MY EMAIL ADDRESSES ARE RUTVA@GMAIL.COM AND RUTVA123@YAHOO.COM. YOU CAN CONTACT ME AT 9876543210. I AM A B.TECH CSE STUDENT LEARNING PYTHON, DATAANALYSIS AND MACHINE LEARNING. PYTHON IS EASY TO LEARN AND PYTHON IS VERY USEFUL FOR DATA ANALYSIS. REGULAR EXPRESSIONS ARE USEFUL FOR EMAIL VALIDATION AND TEXT PROCESSING.

Lowercase Text:

hello, my name is rutva sawarkar. my email addresses are rutva@gmail.com and rutva123@yahoo.com. you can contact me at 9876543210. i am a b.tech cse student learning python, data analysis and machine learning. python is easy to learn and python is very useful for data analysis. regular expressions are useful for email validation and text processing.

Text starts with "Hello": **true**Text contains "Python": **true**

Number of occurrences of "Python": 3

Conclusions: String functions and regular expressions (Regex) are important features of JavaScript that make text processing easier and more efficient. String functions can be used to search, compare, split, replace, trim, and change text, while Regex helps identify and work with specific patterns in the given data. These techniques can be applied to email validation, data extraction, keyword searching, and basic text analysis. When combined with HTML for creating the webpage and CSS for styling, JavaScript provides the required functionality to make the application interactive and user-friendly. Overall, learning string functions and Regex provides a strong foundation for handling text-based data and developing practical web applications.



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Case Study No.: 06

1. **Case Study Title:** To perform basic string operations ; to reverse a string and count number of vowels in a paragraph.
2. **Case Study Program Code:**

```
<!DOCTYPE html>
<html>
<head>
  <title>String Operations</title>

  <style>
    body {
      font-family: Arial;
      background-color: #f2f2f2;
      text-align: center;
      padding: 30px;
    }

    .box {
      background: white;
      width: 550px;
      margin: auto;
      padding: 25px;
      border-radius: 10px;
    }

    input, textarea {
      width: 90%;
      padding: 10px;
      margin: 10px;
    }

    textarea {
      height: 100px;
    }

    button {
      padding: 10px 20px;
      margin: 5px;
      background-color: black;
      color: white;
      border: none;
    }

    .result {
      font-weight: bold;
      margin-top: 15px;
    }
  </style>
</head>
```

```
<body>
  <div class="box">

    <h1>String Operations</h1>

    <!-- Reverse String -->

    <h2>1. Reverse a String</h2>

    <input type="text" id="string"
      | placeholder="Enter a string">

    <br>

    <button onclick="reverseString()">Reverse</button>

    <p id="reverseResult" class="result"></p>

    <!-- Count Vowels -->

    <h2>2. Count Vowels</h2>

    <textarea id="paragraph"
      | placeholder="Enter a paragraph"></textarea>

    <br>

    <button onclick="countVowels()">Count Vowels</button>

    <p id="vowelResult" class="result"></p>

  </div>

  <script>

    // Reverse a string
    function reverseString() {

      let str = document.getElementById("string").value;

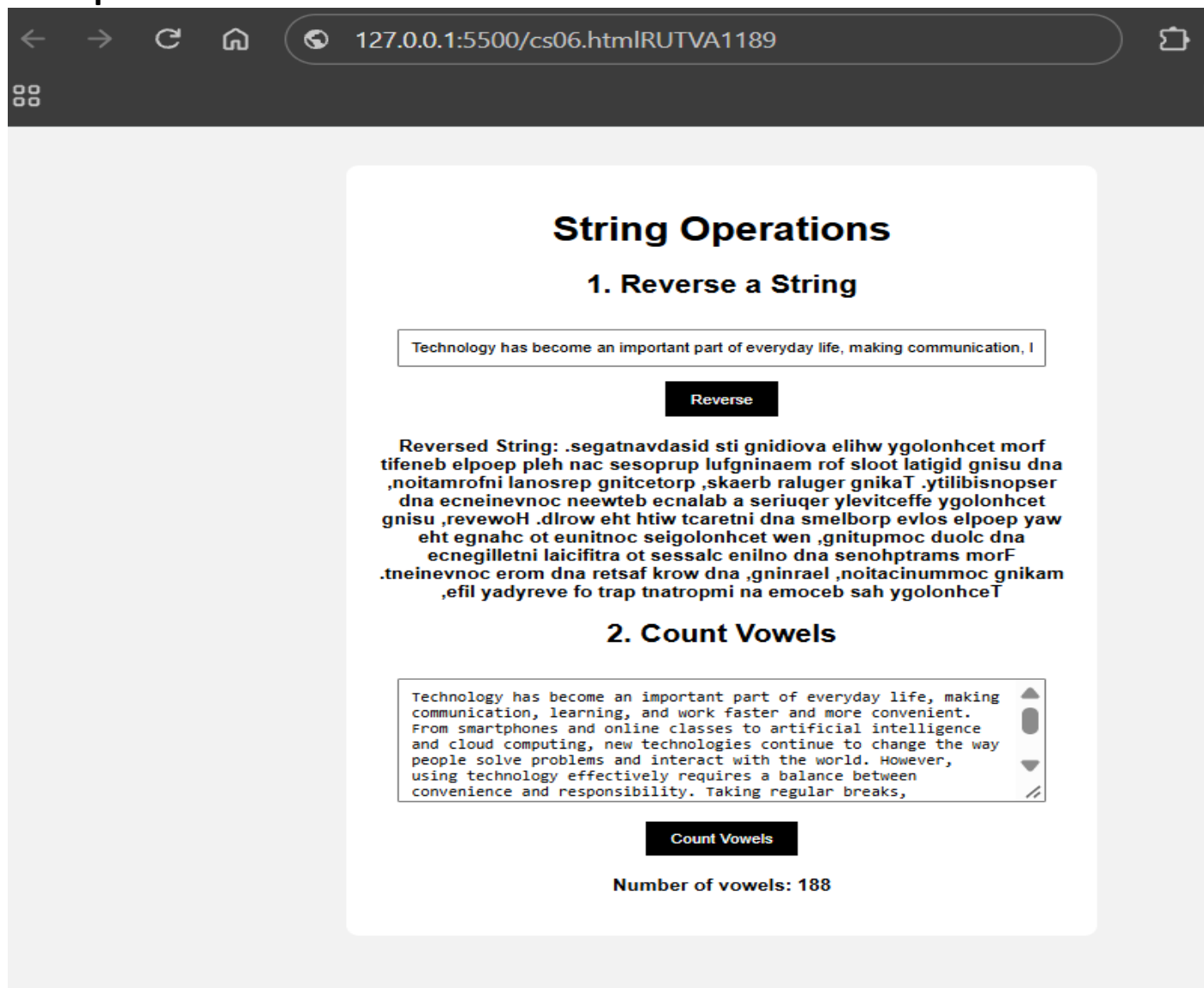
      let reversed = str.split("").reverse().join("");
```

```

91     document.getElementById("reverseResult").innerHTML =
92         "Reversed String: " + reversed;
93 }
94
95
96 // Count vowels
97 function countVowels() {
98
99     let text = document.getElementById("paragraph").value;
100
101     let count = 0;
102
103     for (let i = 0; i < text.length; i++) {
104
105         let ch = text[i].toLowerCase();
106
107         if (ch == "a" ||
108             ch == "e" ||
109             ch == "i" ||
110             ch == "o" ||
111             ch == "u") {
112
113             count++;
114         }
115     }
116
117     document.getElementById("vowelResult").innerHTML =
118         "Number of vowels: " + count;
119 }
120
121 </script>
122
123 </body>
124 </html>

```

3.Output:



4.Result/Conclusion: The JavaScript string operations such as substring, indexOf, split, and replace were successfully performed on the given string. The practical helped in understanding how JavaScript string methods can be used to extract, search, split, and replace text efficiently.